Parkinson’s Disease Presentation

By Maxwell Brimacombe, Ally Loose

PD Quiz

What causes Parkinson’s?

Why can’t you treat with pure dopamine?

Name 3 common symptoms someone with PD may experience

Can you name any common medications someone with Parkinson’s takes?

Name professionals you would expect involved in the care of someone with PD

Have you had a placement or experience where you have worked with someone who has been affected by Parkinson’s?

Subjective Assessment

Presenting Condition

Hidden symptoms:

- Sleep
- Swallow
- Speech
- Fatigue.
- Pain (peripheral neuropathic vs central)

Unusual behaviour / changes in memory or thinking

Bowel/Bladder continence
- High constipation rate

Sexual Dysfunction

Mood
- 40-50% anxiety
- 30% depression

History of Presenting Conditions

1) MDT management:
   - Date of diagnosis?
   - When was their last neurology review?
   - Who is their PD nurse?
   - Previous treatments? Effectiveness?

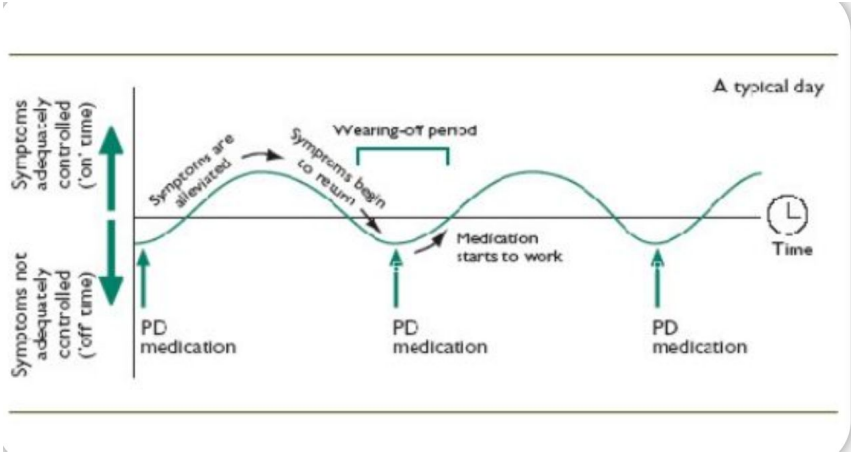
2) Understanding of condition? Goals for treatment?

3) Fall history in past 6 months.
   - Orthostatic hypotension (30-50%)

Drug History

Medications:
Find out their medications, dosage, schedule.
Are they effective?

ON/OFF SYMPTOMS
ON - Effects of medication
OFF - Emergence of PD ssx.



1. "*Wearing off*" is characterized by the re-emergence of Parkinsonian symptoms, usually three to four hours after a dose.
2. *Unpredictable "off"* periods
3. *Freezing of gait (FOG)*.
4. *Failure of the "on"* response.

Medications

TABLE. PHARMACEUTICAL TREATMENTS FOR MOTOR SYMPTOMS OF PARKINSON'S DISEASE

ACTION	DRUGS	AVAILABLE FORMULATIONS	COMMON SIDE EFFECTS
Dopamine precursor with metabolic inhibitor	Levodopa/carbidopa	Tablets (IR, ER) Dissolving tablets	Nausea, vomiting, orthostatic hypotension, vivid dreams, hallucinations, delusions
MAO inhibitors reduce levodopa and dopamine degradation	Rasagiline	Tablets	Hypertension, orthostatic hypotension, potentiation of levodopa-related side effects
	Selegiline	Tablets, capsules, orally disintegrating tablets	
	Safinamide	Tablets	
COMT inhibtors reduce levodopa and dopamine degradation	Entacapone	Tablets	Potentiaiton of levodopa-related side effects, diarrhea, orange color of urine
	Tolcapone	Tablets	Potentiation of levodopa-related side effects, hepatotoxicity,
Dopamine receptor agonists	Pramipexole	Tablets, ER tablets	Nausea, vomiting, orthostatic hypotension, hallucinations, psychosis, impulse control disorders, peripheral edema
	Ropinirole	Tablets, ER tablets	
	Rotigotine	Transdermal patches	
	Apomorphine	Subcutaneous injection	
Other/Unknown	Anticholinergics (trihexyphenidyl, benztropine)	Tablets	Dry mouth, dry eyes, confusion, hallucinations, constipation, urinary retention
	Amantadine	Tablets, capsules, ER tablets	Dry mouth, dry eyes, livedo reticularis, confusion, hallucinations, constipation, urinary retension, peripheral edema

Abbreviations: COMT, catechol-*o*-methyltransferase; ER, extended release; IR, immediate release; MAO, monoamine oxidase.

Social History

ADL screen:
Personal care :
- Feeding
- Dressing
- Washing
- Toileting

House:
- Meal prep
- Cleaning

Work
- Writing
- Phone/computer use
- Shopping

Equipment:
- Walking aids
- Bed levers/support
- Writing/eating implements

Support:
- Package of care / carer
- Support groups for PD.





Mobility

Transfers
- Review combined movements (rolling, turning)

Walking tolerance

Indoor mobility

Outdoor mobility

Falls

Table 2.5.2 Subtypes of Parkinson's disease

Subtype	Associations clinical features
Earlier disease onset (<55 years)	Late onset of falls (approximately 15 years)[223] Late onset of cognitive decline[223] Early onset of freezing (50% after 10 years versus 15 years)[224] Higher risk for anxiety[225] Shorter time to dyskinesias and wearing-off[218] Longer time to Hoehn and Yahr 3[218]
Tremor dominant	Poorer response to levodopa, but slower disease progression[223,226] Lower risk for depression and mood impairments[225] Longer time to and lower risk of dementia compared to PIGD-type[227] Longer time to Hoehn and Yahr 3 compared to PIGD-type[227]
Postural imbalance and gait disorder (PIGD)*	Predominant gait and posture impairments Higher prevalence and severity of depressive symptoms[225,228] Higher prevalence of dementia[226]
Rapid disease progression without dementia	Older age at onset[223] Early depression[223] Early midline motor symptoms[223] In 70% a tremulous onset[223] At post-mortem

Objective Assessment - Ally

Neuro Assessment

ROM / strength / pain / co-ordination / proprioception / sensation / vision

Functional movement and tasks

Common motor findings: bradykinesia, reduced amplitude of movements, rigidity, resting tremor

Restless leg syndrome

Youtube videos

Outcome measures

Hoehn Yahr Scale

The Hoehn and Yahr Scale is the most commonly-used scale to measure the severity of Parkinson's symptoms, and classifies patients in the following stages:[Goetz 2004]

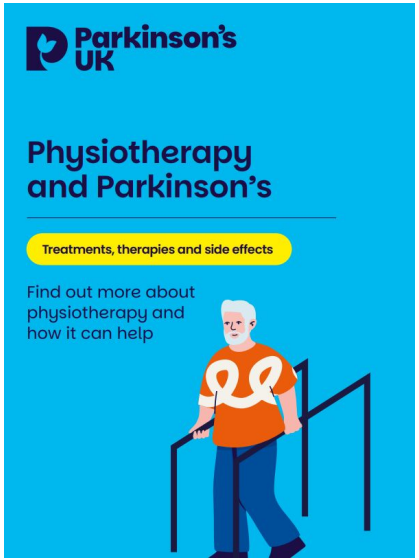
Stage 1	Stage 1.5	Stage 2	Stage 2.5	Stage 3	Stage 4	Stage 5
Unilateral involvement only	Unilateral and axial involvement	Bilateral symptoms No impairment of balance	Mild bilateral disease with recovery on pull test	Mild to moderate disease Physically independent	Severe disability, still able to walk or stand unassisted	Wheelchair-bound or bedridden unless assisted



Outcome Measures - Ally



Rehabilitation strategies



European Physiotherapy Guideline for Parkinson's Disease

Developed with twenty European professional associations

ELSEVIER

Parkinsonism & Related Disorders
Volume 22, Supplement 1, January 2016, Pages S60-S64



Rehabilitation for Parkinson's disease: Current outlook and future challenges

Giovanni Abbruzzese [a b], Roberta Marchese [a b], Laura Avanzino [c], Elisa Pelosin [a b]

Transfers (bed/chair)

- Cueing strategies
- Amplitude training (LSVT BIG / PD warrior)
- Cognitive movement strategies (nose over toes)
- Functional practice (STS)
- Environmental modification

1.7.3 Offer Parkinson's disease-specific physiotherapy for people who are experiencing balance or motor function problems. **[2017]**

1.7.4 Consider the Alexander Technique for people with Parkinson's disease who are experiencing balance or motor function problems. **[2017]**

Kashif *et al. BMC Geriatrics* (2022) 22:381
https://doi.org/10.1186/s12877-022-03035-1

BMC Geriatrics

RESEARCH Open Access

Combined effects of virtual reality techniques and motor imagery on balance, motor function and activities of daily living in patients with Parkinson's disease: a randomized controlled trial

Muhammad Kashif[1,2*], Ashfaq Ahmad[1], Muhammad Ali Mohseni Bandpei[1], Syed Amir Gilani[1], Asif Hanif[1] and Humaira Iram[2]

OM
Modified Parkinson's Activity Scale (M-PAS)
5 x Sit to Stand Test
LPAS

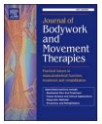
Balance

Journal of Bodywork & Movement Therapies 40 (2024) 520–524

Contents lists available at ScienceDirect

Journal of Bodywork & Movement Therapies

journal homepage: www.elsevier.com/jbmt



The impact of Dance for Parkinson's Disease on subjective well-being and functioning as experienced by dancers and their relatives – A descriptive study

Lars Kristén *, Kristina Ziegert, Peter Karlsson, Elenita Forsberg

School of Health and Welfare, Halmstad University, P.O. Box 823, SE-30118, Halmstad, Sweden

Amaral do *et al. BMC Neurology* (2024) 24:161
https://doi.org/10.1186/s12883-024-03649-5

BMC Neurology

RESEARCH Open Access

Effectiveness of vestibular rehabilitation on postural balance in Parkinson's disease: a systematic review and meta-analysis of randomized controlled trials

Carla Marineli Saraiva do Amaral[1], Samuel Brito de Almeida[1], Renata Parente de Almeida[2], Simony Lira do Nascimento[3], Rodrigo Mariano Ribeiro[1] and Pedro Braga-Neto[1*]

Journal of Physical Medicine Rehabilitation Studies & Reports

SCIENTIFIC Research and Community

Review Article Open Access

The Science Behind Balance Training in Physical Therapy: Preventing Falls through Neuromuscular Reeducation

Shivangi Chaughule

Movement disorders

OPEN ACCESS

Original research

Effect of long-term Tai Chi training on Parkinson's disease: a 3.5-year follow-up cohort study

Gen Li [1], Pei Huang,[1] Shishuang Cui,[1] Yachao He,[1] Yuyan Tan,[1] Shengdi Chen [1,2]

OM
Timed up and Go
BERG balance
Tinnetti
Push & release
mCTSIB

Gait

Neurorehabilitation and Neural Repair
Volume 32, Issue 10, October 2018, Pages 872-886
© The Author(s) 2018, Article Reuse Guidelines
https://doi.org/10.1177/1545968318801558

SAGE journals

Review Article

Exercise Guidelines for Gait Function in Parkinson's Disease: A Systematic Review and Meta-analysis

Meng Ni, PhD[1], Joseph B. Hazzard, EdD[1], Joseph F. Signorile, PhD[2,3], and Corneliu Luca, MD, PhD[3]

Movement Disorders
CLINICAL PRACTICE

VIEWPOINT

MDCP CONFERENCE ON UNMET NEEDS AND UNANSWERED QUESTIONS IN CLINICAL PRACTICE – LONDON, 2021

Compensation Strategies for Gait Impairments in Parkinson's Disease: From Underlying Mechanisms to Daily Clinical Practice

Anouk Tosserams, MD,[1,2,] Bastiaan R. Bloem, MD, PhD,[2] and Jorik Nonnekes, MD, PhD,[1,3]*

OM
Dynamic Gait Index
Functional Gait Assessment
M-PAS Gait

Freezing of gait - MAX

Freezing of Gait

- Example Freezing of gait video

  - Cloth coaster
  - Tape marks

MUSIC!

Beats Per Minute (Music Database :: Dave Tompkins)
- External auditory cueing
- Snapping, metronome, counting (1234,1234)





How does Physiotherapy intervention change over time?

Fig 4.1. Core areas of physiotherapy related to disease progression

